Hyper-**N**etwork for **electro-mobility**

# NeMo D7.1. Global adoption methodology and self-certification methodology for CPOs and EMP

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# Glossary of Terms

| Term | Description |
|---|---|
| Electro mobility or e-mobility | The use of electric-powered drivetrains for road vehicles designed to shift vehicle design away from the use of fossil fuels and carbon gas emissions. |
| e-roaming or eRoaming | A market model in electro mobility whereby EV drivers may charge their vehicles at all charging stations, regardless of any contracts concluded with operators. The billing then takes place via the customer's own contractual partner. |
| Hyper-Network | A distributed environment with open architecture based on standardised interfaces, in which actors (physical or digital) can connect and interact seamlessly, exchange data and provide integrated and interoperable ICT services. |
| Hyper-Network Business Partner | A user of the Hyper-Network which NeMo or the BAEM has accredited as being compliant with the relevant protocols and procedures. Potentially a "label" that could be used by the organisation concerned. |

# List of abbreviations and acronyms

| Abbreviation | Meaning |
|---|---|
| BAEM | Business Alliance for Electric Mobility<br>*A proposed governance structure for the Hyper Network after the end of the NeMo project, that will be a legal entity composed of members* |
| CIM | Common Information Models |
| CP | Charge Point |
| CPO | Charge Point Operator |
| DSO | Distribution System Operator |
| EM(S)P | Electro-mobility (service) Provider |
| eMI³ | eMobility ICT Interoperability Innovation group |
| EV | Electric Vehicle |
| ICT | Information and Communication Technologies |
| IEC | International Electrotechnical Commission |
| ISO | International Organization for Standardization |
| ITS | Intelligent Transport Systems |
| OEM | Original Equipment Manufacturer (automotive industry) |
| SoC | Status of Charge |
| SoH | State of Health |
| VS | Virtual Sensor(s) |

# Executive Summary

This deliverable proposes a global adoption methodology that allows service providers to evaluate the quality of data and service in the NeMo Hyper-Network and a strategy to attract new actors in it. Furthermore, the deliverable identifies a set of indicators for self-certification and proposes a business model for an external certifier to ensure quality of data and services.

The first part of the deliverable is dedicated to the analysis of the existing literature on data and service quality with the aim to understand how data quality indicators are classified and how quality of services is measured. Based on the analysis of existing studies, several dimensions and characteristics of data and service quality have been selected based on their relevance to the NeMo services. The second part of the deliverable is dedicated to the identification of existing services similar to NeMo. In this way, it is possible to identify potential actors who could be interested to be part of the NeMo Business Alliance for Electro-Mobility (BAEM).

The second part of the deliverable is dedicated to the description of the data deployed by each NeMo service and to the definition of the requirements to ensure that service and data quality are provided. Each service is initially described in terms of type of data needed, type of service and information provided. Based on the high-level description of NeMo services, a set of indicators to measure the NeMo services' quality level are described by also considering the possibility to carry out the evaluation in the context of the validation activities carried out in the project. After identifying the data and quality standards for each NeMo service, the deliverable proposed methodologies for self-certification of services by service providers. Finally, the last part of the document defines a business model framework to ensure that quality levels are maintained in the Hyper-Network during and after the execution of the project and presents a process to attract new Actors in the Hyper-Network.

# 1. Introduction

Lack of interoperability in terms of electromobility services, due to high number of involved stakeholders and scarcity and variability of data makes difficult for the EV drivers to plan and have charging sessions during a long trip via a seamless experience. The vision of NeMo project is to develop a Hyper-Network in order to facilitate the connection among all service providers and service consumers based upon tools and models so that all involved actors can exchange data and information using the same language on a decentralised and secured network. Thanks to the NeMo Hyper-Network, the final users can easily access the infrastructure while new business opportunities can emerge. The Hyper-Network is based on a Common Information Model, creating and using data translators to convert data structures, the Inter-roaming protocol contributing to the Pan-European eRoaming Framework, a set of innovative Horizontal services, an open cloud market place where services and business information are shared, and a self-certification process for ensuring quality of provided services.

This Deliverable analyses the existing approaches for data and service quality evaluation and proposes the most appropriate approach for the quality assessment of the NeMo services. The first objective is to define a methodology for quality self-evaluation that consists of a set of quality indicators for self-certification, as well as suggest a business model for an external certifier, who takes charge of evaluating new and existing services provided in the NeMo Hyper-Network. The definition of a certification process is needed to ensure a high-quality level of NeMo services even after the conclusion of the project. The second objective is the analysis and development of a strategy for the adoption of the NeMo specifications and requirements by external stakeholders, aiming to define a strategy to increase and extend the participation of new users by creating a network that is self-sustainable.

The document is structured in four main sections. The first section is dedicated to data and service quality based on the analysis of existing studies and projects. The overview of how services are evaluated allows for the identification of indicators that are important for quality assessment of NeMo services. The second section presents a set of existing services similar to those presently provided through NeMo, which is important when targeting inclusion in the NeMo Hyper-Network of new service providers. The third section presents a detailed description of the data and service quality indicators for each NeMo service. This allows to identify specific measures that are successively compared to those identified in the literature review analysis, and to finally identify a restricted set of general indicators that can be taken as a reference when the service provider wants to perform a self-certification. The final section presents the results of a questionnaire survey that was initiated during the 2nd NeMo Stakeholders Forum Conference, held in Brussels on 9 October 2018. The survey addressed the definition of the most functional and acceptable contractual model, between the service providers, the consumers, the NeMo Business Alliance for Electro-Mobility (BAEM) which is the association that will be responsible for the Hyper-Network and new actors potentially interested in taking part in NeMo, in order to guarantee the maximum participation from external stakeholders in the future.

# 2. Data and service quality

Chapter 2 provides an overview of existing studies on data quality and service level evaluation. This general overview was the basis for the definition of data and service level that should be ensured when NeMo services are provided. The literature provides different classification of data quality indicators. Since the characteristics of data quality are highly dependent on the sector considered, the objective of this section is to present a selection of works that provide a general classification of data characteristics and approaches in order to determine the quality characteristics of NeMo services. Secondly, this section provides an overview of existing methodologies to evaluate service quality level with a focus on IT services that provide information to users and on approaches that could be considered for evaluation NeMo services quality.

## 2.1 Data Quality

Data quality can have tremendous impact on the businesses themselves as well as the services provided. Although of such importance, there is no common agreement on the definition of data quality, because it is highly dependent on the context in which data are needed. Generally, data quality is important when services consist of information provided to the users, such as in the case of NeMo. In the scientific literature, several works propose definitions and classification of data quality characteristics. The problem is so wide that there is an annual conference organized since 1996 by Massachusetts Institute of Technology (MIT) on this specific topic, the International Conference on Information Quality (ICIQ). Furthermore, the evaluation of data quality is sector-dependent (i. e. the characteristics of data quality in the health sector are completely different compared to those in the transport sector). According to ISO/IEC 25012 standard, data quality is "the degree to which data satisfy the requirements defined by the product-owner organization".

| Data quality indicators | ISO/IEC 25012 | Wang, Strong (1996) | Cykana, Paul, Stern (1996) | Wand, Wang (1996) | Redman and Blanton (1997) | Larry (1999) | Naumann and Rolker (2000) | Redman (2001) | Pipino, Lee, Wang (2002) | Long and Seko (2002) | Lee, Strong, Kahn, Wang (2002) | Bovee, Srivastava, Mak (2003) | Stvilia, Gasser Twidale, Smith (2007) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Completeness | X | X | X | X | X | | X | X | X | | X | X | X |
| Consistency | X | X | X | X | X | X | X | X | X | | X | X | X |
| Timeliness | | X | X | X | | | X | X | X | X | X | X | |
| Accuracy | X | X | X | X | | | X | X | | X | | X | X |
| Accessibility | X | X | | | | | X | X | X | X | X | | X |
| Relevancy | | X | | | X | | X | X | X | X | X | | X |
| Understandability/ clarity | X | X | | X | X | | X | X | X | | X | | |
| Interpretability | | X | | X | | | X | | X | X | X | | |
| Security | | | | | | | X | X | X | | X | | X |
| Value added | | X | | | X | X | X | | X | X | | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Credibility/ believ-ability | x | x | | | | | x | | x | | x | | |
| Conciseness | | x | | x | | | x | | | | x | | |
| Objectivity | | x | | | | | x | | x | | x | | |
| Reputation | | x | | | | | x | | x | | x | | |

Table 1 presents a summary of the most common data quality indicators presented in literature. Overall, there is a general agreement to define data quality in terms of five characteristics: **completeness**, **consistency**, **timeliness**, **accuracy** and **relevancy**.

| Data quality indicators | ISO/IEC 25012 | Wang, Strong (1996) | Cykana, Paul, Stern (1996) | Wand, Wang (1996) | Redman and Blanton (1997) | Larry (1999) | Naumann and Rolker (2000) | Redman (2001) | Pipino, Lee, Wang (2002) | Long and Seko (2002) | Lee, Strong, Kahn, Wang (2002) | Bovee, Srivastava, Mak (2003) | Stvilia, Gasser, Twidale, Smith (2007) | Loshin (2011) | Schaal, Smyth, Mueller, MacLean (2012) | UNECE/HLG | Cappiello et al. (2004) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Completeness | x | x | x | x | x | | x | x | x | | x | x | x | x | x | x | x |
| Consistency | x | x | x | x | x | x | x | x | x | | x | x | x | x | x | | x |
| Timeliness | | x | x | x | | | x | x | x | x | x | x | | x | x | x | x |
| Accuracy | x | x | x | x | | | x | x | | x | | x | x | x | x | | x |
| Accessibility | x | x | | | | | x | x | x | x | x | | x | | x | x | x |
| Relevancy | | x | | | x | | x | x | x | x | x | | x | | x | | |
| Understandability/ clarity | x | x | | x | x | | x | x | x | | x | | | | x | | |
| Interpretability | | x | | x | | | x | | x | x | x | | | | x | | x |
| Security | | | | | | | x | x | x | | x | | x | | x | x | |
| Value added | | x | | | x | x | x | | x | x | | | | | x | | |
| Credibility/ believability | x | x | | | | | x | | x | | x | | | | x | | |
| Conciseness | | x | | x | | | x | | | | x | | | | x | | |
| Objectivity | | x | | | | | x | | x | | x | | | | x | | |
| Reputation | | x | | | | | x | | x | | x | | | | x | | |

Table 1: Comparison between different data quality categories of indicators

The classification of data quality represents the first step towards the evaluation of quality of data. *Laranjeiro et al. (2015)* surveyed the state of art on data quality classification, dimensions, and main problems. According to the authors, the main problem related to data quality is the lack of standardization and the availability of a huge variety of definitions. Based on the literature, they classified data quality in four categories: **Intrinsic**, **Contextual**, **Representational** and **Accessibility**.

Another example of data quality classification is presented by *Cappiello et al. (2004),* who showed how data quality assessment should take into account users' requirements variation with the accessed service. This paper underlines how the data quality literature classifies data quality dimensions as **accuracy**, **completeness**, **consistency**, **timeliness** (currency, volatility), **interpretability** and **accessibility**. The model presented in this paper approaches data quality from the user perspective. Since the authors recognise that the same data can be judged as satisfying or poor according to the expectations and needs of different users, in the proposed model, users with similar characteristics are classified based on the set of services they access. The model architecture is, therefore, made up of three modules: the selector module, the quality assessment module and the profiling module.

In *Batini et al. (2009),* the authors provide a description of the main methodologies that help the selection, customization and application of data quality assessment. Several dimensions are considered to compare those methodologies in order to create a set of guidelines that can define an objective process to assess the quality of data. In particular, the authors identified some strategies that are common in many methodologies. In general, the sequence of activities of the data quality methodologies analysed are the following:

- state reconstruction (collection of contextual information);
- assessment / measurement (measurement of the quality of data collections using quality dimensions)
- improvement (all those steps and strategies useful to reach new data quality targets).

The assessment phase is consistent among the methodologies taken into account by the authors. Data quality assessment generally consists of the following phases: data analysis, data quality requirement analysis, identification of critical areas, process modelling, and quality measurement. While the most common steps of the improvement phase are: costs evaluation, process responsibilities assignment, data responsibilities assignment, identification of the causes of errors, selection of strategies and techniques, design of data improvement solutions, process control, process redesign, management improvement, monitoring improvement. Furthermore, the authors discovered that most of the research agreed on the basic set of data quality dimensions: accuracy, completeness, consistency and timeliness. Finally, the authors divided the analysed methodologies in four categories: complete methodologies, audit methodologies, operational methodologies and economic methodologies.

Data quality assessment has been analysed in terms of the type of problems that may characterize data. In *Oliveira et al. (2005*), a formal definition for each data quality problem and a taxonomy of those problems are presented. The taxonomy is based on the well-known model of data organization (data is stored in a multiple data source, which is composed of several relations among them, each relation is made up of several tuples and each tuple is composed by a certain number of attributes). The identified problems can be represented in several levels:

**Data quality problems at the level of attribute/tuple**. This level is divided in three more specific groups of data quality problems:

- problems that were encountered by analysing the values: missing value, syntax violation, incorrect value, domain violation, violation of business domain constraint, invalid substring, misspelling error, imprecise value.
- single attribute of a single tuple: unique value violation, existence of synonyms, violation of business domain constraint.
- single attribute in multiple tuples: semi-empty tuple, violation of functional dependency, violation of business domain constraint.
- multiple attributes in single tuples: semi-empty tuple, violation of functional dependency, violation of business domain constraint.

Data quality problems at the **level of a single relation**. The data quality problems of this level are: approximate duplicated tuples, inconsistent duplicated tuples, violation of business domain constraints.

Data quality problems at the **level of multiple relation.** Problems that appear in analysing the values from multiple relations: referential integrity violation, incorrect reference, heterogeneity of syntaxes, circularity among tuples in a self-relationship, violation of business domain constraint.

Data quality problems at the **level of multiple data sources.** This group includes problems that were identified thanks to the analysis of the values of multiple data sources, which are: heterogeneity of syntaxes, heterogeneity of measure units, heterogeneity of representation, existence of synonyms, existence of homonyms, approximate duplicate tuples.

Besides the identification of the specific characteristics of data quality, a certification methodology should be based on a set of indicators that are common for all NeMo services. The identification of these indicators is the result of the cross-analysis between the literature review indicators and the specific indicators identified for each NeMo service later in Chapter 4.

## 2.2 Service Quality

The importance of **service quality**, along with the customer switching behaviour, is among the most important factors for the companies' profitability. In *Lapierre et al. (1996),* the authors present the most important studies on the empirical work on service quality in order to help researchers and managers measure service quality in the best possible way. Moreover, the authors underline that the final goal of properly evaluating the service quality is to improve customer satisfaction and relations. This research showed that the two most important schools of thoughts about quality evaluation are the Nordic and the American ones. An interesting part of this paper was dedicated to the difficulties met when evaluating service quality. In particular they presented four questions that needed an answer:

- what should be measured?
- which service quality perception should be evaluated?
- should the emphasis be on the technical quality? Should it be on functional quality? Should it be on both?
- should the service dimensions be specific? Should it be generic? Should it be both?

In *Caruana et al. (2000),* the authors are interested in understanding whether the **value of a good/service plays a role between service quality and satisfaction**. In fact, many experts are confusing three important concepts: service quality, customer satisfaction and value. For this reason, the authors developed a model to measure these concepts. A questionnaire made up of 20 items was prepared: 16 items were dedicated to the service quality, three items to the satisfaction and 1 item to the value. 80 people were interviewed over a period of 4 weeks. The results showed that while satisfaction, value and service quality are correlated, they are also clearly distinct from each other. The results of this study also confirmed a **direct connection between service quality and satisfaction**, suggesting that the effect of quality on satisfaction is moderated by value of good/service. Finally, the results show that even if buyers provide high feed backs on levels of service quality, it is not granted that their satisfaction will also be high: **prices** still have an important role on customers' satisfaction.

*Cronin and Taylor (1992),* assume that managers of service providers need to know **how to measure service quality** and the aspects that define quality and satisfaction. Starting from these assumptions, the authors demonstrate that the SERVQUAL scale is inadequate and propose an alternative solution. They also understand the relationship among service quality, customer satisfaction and purchase intentions. Although this relationship has been theoretically studied, in fact, it has never been empirically tested. Regarding the first point, the authors of this research propose a new performance-based scale, the **SERVPERF,** and show how it is more efficient compared to the SERVQUAL scale. For instance, it reduces from 44 to 22 the items to be measured. Regarding the second point, the results highlight that **the quality of the service comes before the satisfaction** and that customer satisfaction exerts a very strong influence on the purchase intention (much more than quality). Furthermore, the results show that, most of the time, buyers do not aim for the highest quality of service as the price and availability are enough to increase overall satisfaction. The research results also showed that the **items that define the quality of services in a sector are not necessarily valid for all sectors** (the authors hypothesize that we can distinguish between low involvement services and high involvement services).

*Hung el al. (2003),* point out that customer satisfaction (customer perception) and the importance that the customer gives to a service / product (customer expectation) are the most important factors for determining service quality performance. A customer satisfaction index and a customer expectation index are then proposed. Based on the Performance Evaluation Matrix developed by Lambert and Sharma, the authors proposed a Service Quality Performance Matrix with the aim of standardizing and simplifying the service quality assessment procedure.

**Measuring Information Technologies service quality**

The problem of measuring the service quality has been recently tackled with reference to services offered through the use of new technologies such as internet and mobile phones. The evaluation of service quality delivery through web sites has been analysed by *Zeithaml et al (2002)*, resulting that academic researchers, managers and marketing experts have all identified a number of important criteria that consumers use in evaluating the quality of the service on a website. Although the multitude of criteria proposed in various research, there are currently no common criteria recognized by all. The most frequently accepted and used ones are: **fulfilment, privacy / security, site design, efficiency and ease of use.** The need to have more than one criterion to identify the quality of the service, shows how the electronic service quality is multifaceted. Another interesting aspect that this paper underlines concerns the fact that the e-SQ does not affect only the satisfaction but also the intent to buy, and the actual purchase. Finally, this work pointed out that there is still a long way to go in terms of analysing service quality delivery through web sites, and that it is fundamental to globally analyse the antecedents, the composition and the consequences of a certain service quality.

*Zemblytė (2015)* based the definition of *Zeithm, Parasuraman and Malhotra (2000)* which sees e-service quality as "the extent to which a web site facilitates efficient and effective shopping, purchasing and delivery", develop and test the tool to evaluate e-service quality through an intense literature review, an online survey and a statistical analysis of data.

The online survey was developed in 3 different sections:

- the first was dedicated to gathering information regarding the experiences of respondents connected to e-service usage;
- the second was dedicated to collecting information concerning e-service quality;
- the third included questions on demographic characteristics.

The results of this research show that according to the customers' perception the e-service quality is a four-dimensional construct: **compensation**, **response** and **fulfilment**, **website operation** and **reliability**.

Starting from the idea that, since the Internet coming into everyday life has drastically changed not only the behaviour of people but also the **decision-making process** when buying a product, *Palese and Usai (2018)* analysed the service quality measurement problem using socialized data. The aim was to offer important insights for the designers of online review systems and for those involved in market analysis, seeking new approaches to **measuring service quality**. Referring to the SERV-QUAL model that was proposed by *Parasuraman et al. (1985)* as a theoretical basis, this research focuses on a price comparison website, designed to make (almost) impossible fake reviews: the reviews available for this study were 27.117, collected from the site during a period of 8 years. The results demonstrated that **socialized textual data is able to provide a wealth of customer service information that is appropriate for measuring service quality.** Furthermore, the authors have shown that the distribution of topics in online opinions differs between positive and negative reviews.

*Liang et al. (2013)* identify the critical factors that push the people in switching mobile phone service providers; they carried out a survey on 400 customers and they discovered seven critical factors: **the core service failure**, **the high price**, **the ethical problems**, **the competition**, **the inconvenience**, **the service that encountered a failure**, and the **influence from the user's group**. It is important to underline that this study was carried out in China, therefore it would be ideal to use the same methodology and apply it to other countries.

In *Huang (2015)*, the author assumed that there are many mobile services available but that the ideal characteristics for these services are not yet known. By conceptualizing, constructing, refining and testing a M-S-qual (a multiple-item scale), this research aimed to **develop a scale to measure service quality in the mobile environment**. A questionnaire and an evaluation scale were then prepared. Starting from the framework proposed by *Hinkin (1998)* (Item generation, questionnaire administration, initial item reduction, confirmatory factor analysis, convergent / discriminant validity and replication), there were therefore 51 items that covered **9 factors** that were used to measure the mobility service quality construct: **efficiency, system availability, contact, privacy, fulfilment, responsiveness, compensation, content, billing**. Thanks to the model proposed in this study, mobile service providers will be able to assess the quality of their services and, at the same time, researchers will be encouraged to develop theories related to mobile service quality.

Based on the analysis of selected works, the quality of NeMo services could be measured using a scale to evaluate the **efficiency**, **system availability**, **reliability** and **easiness to use**.

# 3. Existing services

This section presents an overview of existing services that have similar characteristics to those developed in the context of NeMo project. The objective is to identify the functionalities that are already in the market or have already been developed, and the Actors that might be interested in becoming part of the BAEM.

## 3.1 Existing services similar to NeMo Load forecasting due to EV charging

The **Status of charge battery measurements using data accumulation**[1] is a patent, issued by David Elder and William Weiss, which aims to improve the accuracy of the state of health and charge calculation of a battery. The patent describes a system that has the goal of tracking and monitoring the performance data of batteries, including this information in the state of health (SoH) and/or the status of charge (SoC) calculations. The main characteristics consist of the possibility to develop a system (composed by a database, a computer network, a point of sale/point of maintenance device, and an electronics package on a battery product) that stores information on the battery and, therefore, extract important and useful information to improve SoH or SoC algorithms or methods and then update these on a battery product.




The **myEV electric vehicle logger & app** is an application developed by MyCarma[2] that consists of an EL-Vs logger, which tracks energy consumption and the status of charge (SoC) on several production plug-in vehicles. Thanks to the feedback from the users and the data from the logger, the service can provide information that reduce range anxiety. From the app which is shown in Figure 1, the users are able to: get real-world data on the trip's efficiency, get information about the battery health level, make a comparison with users close to each other. myEV has also developed a public Charging Etiquette that allows, by placing a personalized QR code on a window of the car, other EL-Vs users to ask if they can unplug their vehicle and free the charging station.

Figure 1: myEV app

Fleetcarma[3] is a logger that can provide real-time vehicle information and charging data. In this way, the driver can receive the vehicle's battery status of charge, so that range anxiety is lowered. Fleetcarma also offers the possibility to transform the data in statistics to understand charging behaviour (see Figure 2). Information on charging behaviour is important for charging point operators.

[1] patents.google.com/patent/WO2010024892A1/en
[2] www.mycarma.com/products/myev/
[3] www.fleetcarma.com/smartcharge-manager/

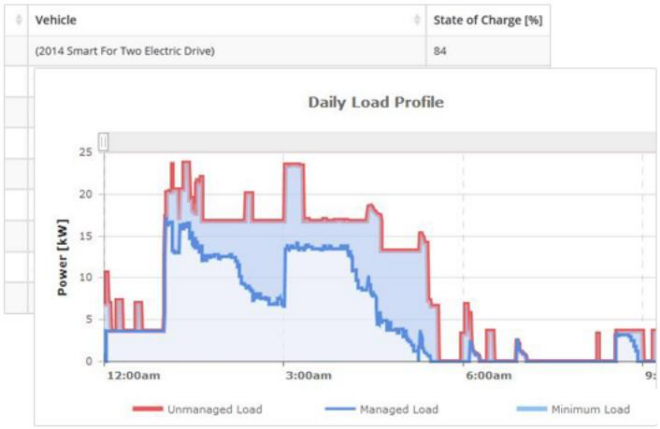



Figure 2: Fleetcarma smart charging programme

## 3.2 Existing services similar to the NeMo Thermal Battery Pre-conditioning

One of the main problems of electric vehicles is the cold climate since it creates losses in the electric vehicle range. Moreover, extreme cold can limit the performance and efficiency of the battery. For this reason, **Tesla** has developed a **new mobile app update with battery pre-conditioning feature,** which consists on a battery management system that aims to warm the battery. Similarly, BMW has added to its App BMW i3 a function that allows the preconditioning of the electric vehicle. Through the app, the user can set the time in which both the cabin and the battery will be warmed. The result is a better ride both for the battery and the driver. If the car has been preconditioning while it was still plugged in, the range increases. Finally, Ford Global Technologies, proposed a system[4] that aims to precondition the battery of a hybrid electric vehicle when it detects cold temperature.

## 3.3 Existing services similar to NeMo Battery capacity calculation

The capacity calculation service is provided by Ev-Move[5] that is an application provided by the company BeNomad. The company provides several services for fleets' management such as the possibility to estimate the energy cost for a certain route. This calculation takes into consideration several factors such as the SoC, the characteristics of the road, the driver's profile, the vehicles characteristics, the weather conditions, and the energy recovery capacity. Other features of this app are the possibility of searching a charging station and planning a route.

[4] patents.google.com/patent/US6271648
[5] benomad.com/en/products/business-solutions/ev-move/

## 3.4 Existing services similar to NeMo Battery Management

Battery Management systems are usually outsourced and/or procured on the market by automotive industry. The **Battery Management System**[6] provided by Texas Instruments, for instance, helps to monitor and control the high-voltage battery stack. These features allow to measure the cell's charge, voltage, temperature, and health, to have data that allow diagnosis and protection for the battery and to guarantee cell balancing. Similarly, Renesas[7] has created several automotive grade Li-ion battery management solutions that were specifically designed to meet performance, safety, reliability requirements of the next generation of electric vehicle applications. Finally, Elthion[8] produced the Lithiumate Pro BMW that was designed in order to interface with several motor controller and chargers. Moreover, it can be used to manage a traction pack intelligently.

In the market, there are also applications such as Electrip[9] that provides the possibility of planning a trip with optimized charging steps, the maximum range calculator that keeps adjusting according to the condition of the trip (driver's style, road peculiarities, traffic…), the integrated turn-by-turn navigation, the possibility of rating and review the charging stations, an accurate prediction on how long it will take for the driver to arrive to the desired destination and the possibility to track the battery performance in order to compare it with other electric vehicles.

## 3.5 Services aimed to reduce EV Range Anxiety

Range anxiety is one of the main barriers for EV drivers. However, the market is full of products such as applications and tools that offer various services for EVs:

The **Has To Be**[10] is a company that provides several e-mobility products and services with the goal of facilitating the adoption and the use of EL-Vs. Some of the offered e-mobility services are:

- chargepoint operation, a service that ensures the optimal operation of the charging infrastructure (i.e. 24/7 monitoring remote trouble sheeting);
- eMSP.Operation, a service that allows drivers to charge everywhere in Europe with no limits;
- eDriver.HOTLINE that guarantees a professional customer service, giving instant support (24/7) at the charging station;
- eFleet.PAYBACK, a service that enables employees to charge the company cars and get reimbursed easily.

[6] www.ti.com/solution/battery_management_high_cell_count_for_hev#referencedesigns
[7] www.renesas.com/eu/en/products/automotive/battery-management-systems.html
[8] www.elithion.com/automotive_battery_management_systems.php
[9] electrip.co
[10] has-to-be.com/en/

The **PlugShare**[11] has the goal of accelerating the adoption of EL-Vs by unifying drivers and industry. With Plugshare (available also as an app) the user can find charging stations in Europe (see Figure 3), Middle East, North Africa, North-central America and Oceania. With Plugshare the user can add several filters to find the best solution for his/her vehicle and it is possible to plan a specific trip by considering the location of charging stations. Furthermore, the user has access to the Plugshare Store, a free service that aims to offer several tools for electric vehicles.

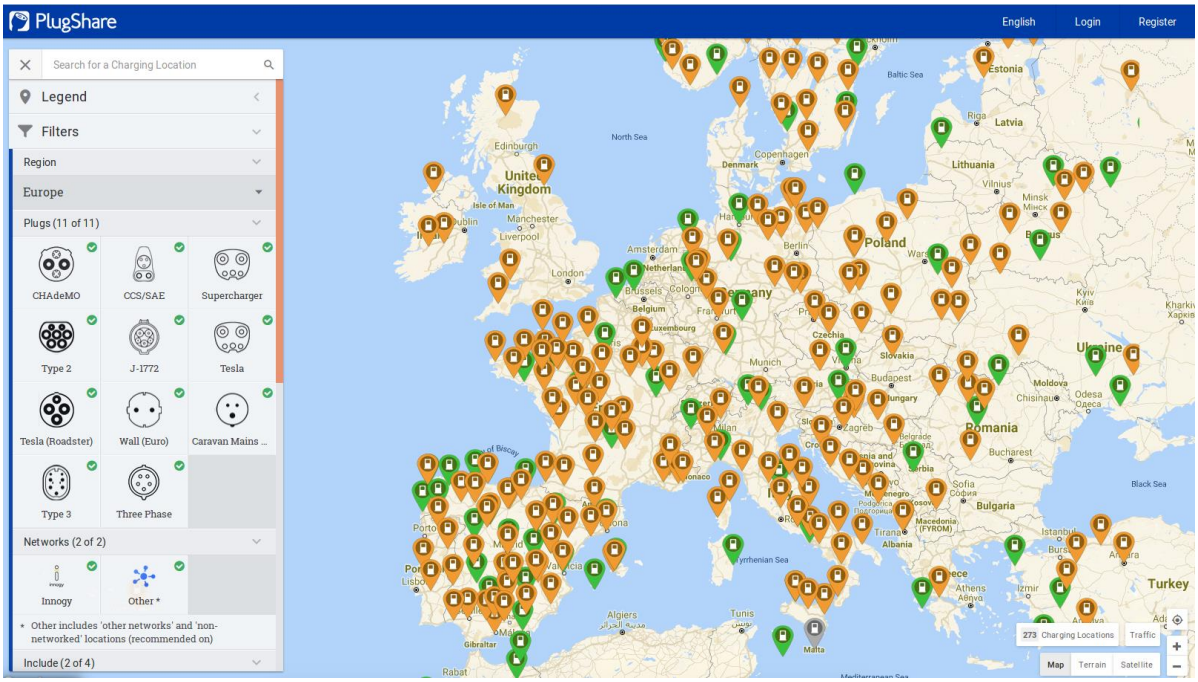



Figure 3: PlugShare

Finally, Plugshare, in collaboration with Go Electric Station and Next Charge, adds the possibility to be part of a community of users, who want to provide their own private outlet. In this way, the users will be able to know the availability of public charging systems and the ones of the private outlets.

The **Charge Hub calculator**[12] allows a user to calculate how much money he/she would be able to save if he/she would own an electric car (Figure 4). With the Charge Hub Map the user can find all the available charging stations in USA and Canada, plan the ideal trip so that the user can recharge his/her electric car at the perfect stop according to his/her route and destination. Furthermore, Charge Hub has created a very completed guide (Charge Hub Guides) in which several topics related to electromobility are considered. Among others, the guidelines include the right use of the battery charging at home and the incentives available according to country and province. Finally, Charge Hub has developed the Charge Hub store, where it is possible to compare and order models of several products useful for the electric vehicles.

[11] www.plugshare.com
[12] chargehub.com

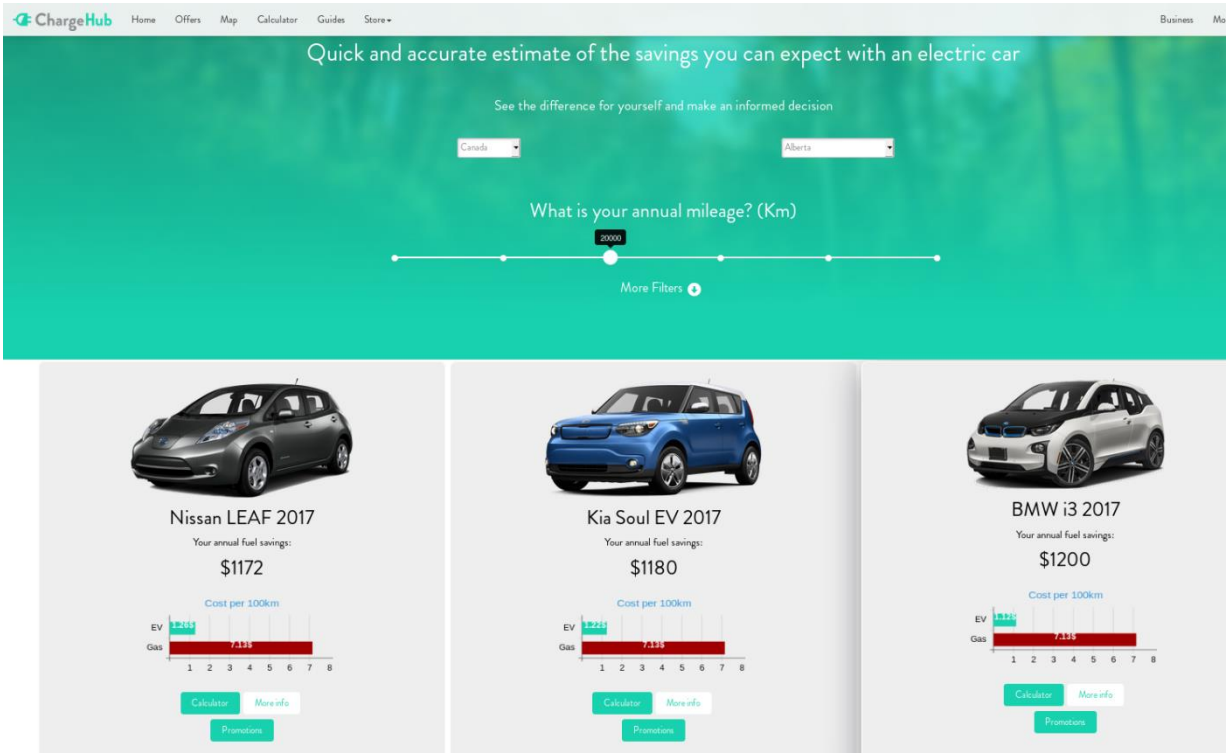



Figure 4: Charge Hub calculator

**NextCharge**[13] is a system that focuses on the easiness of payment. In fact, with Nextcharge the user can check the different prices, choosing among several ways of payment and he/she can check his/her payment directly from the app. With Nextcharger, the user has also the possibility of using the "Nextcharge Wallet", an electronic wallet that allows to check the expenses related to the charging of a specific car (Figure 5).




Figure 5: NextCharge application

[13] goelectricstations.it

**E-MOVING**[14] is a project developed by A2A which aimed to install 100 charging boxes between Brescia and Milan (Italy). The car can be charged in one hour and it provides an app that offers a map useful to find the best charging point according to needs.

**CHARGEBUMP**[15] focuses on the problem of charge point occupancy. The app provides the possibility to send a message to the driver, who is occupying the charge point by indicating the plate number. In this way, it is possible to leave the vehicle while it is charging with no pressure to move it as soon as it finishes to charge. This app is based on the idea that instantaneous messaging between drivers can be a quick and nice solution to avoid stressful situations related with electric vehicles.

**Charge Map**[16] provides a very well updated map of all the charging stations available in Europe and in USA. Chargemap is also an app that allows the user to activate several filters in order to find the best option for his/her needs. Each station has a very detailed sheet and the app proposes the driver the best route for his/her electric trip. Furthermore, this is a collaborative app: it is supported by a very active community of European electric vehicle drivers. The Community consists of about 200.000 members, 200.000 photos and 300.000 reviews and comments.

**EVWAY (Route220)**[17] created a map in which the user can find all the charging stations close to him/her and, thanks to a deep description of each station, the user will be able to choose the best structure for him/her and his/her car's needs. In this app, the driver can also discover hotel and restaurant with an available charging station.

The service **E-Go**[18] proposed from the ALD automotive is based on the idea that renting a car (and in particular an electric car) is the best solution both for private and business. For this reason, they created E-go, the 100% electric solution that allows drivers to move freely around the city (no LTZ for ELV-s, no restriction of any kind), spending less money and taking care of the environment. In order to offer more services for the electro mobility, there is an app (My E-go) that allows the users to better control their car.

**Enel X Box Station** allows the user to rent not just a car, but also a box station to put in the users' private parking. The service includes a map with all the charging station available and updated research and materials to inform about the newest technology and the new ways of charge.

**EQ Control**[19] is the app developed by Smart to have an overview of the charging status, based on the autonomy and the residual charging time. Moreover, this app plans the best moment to charge the vehicle according to the day/night current rates. Also, the EQ control app allows to check stations availability along the route and it allows to pay charge directly from the phone. Other services are:

- Analysis of the driving style (with a consequent proposition of an ideal driving style);

[14] www.e-moving.it/home/cms/emv/
[15] chargebump.com
[16] chargemap.com
[17] evway.net
[18] mobilitysolutions.aldautomotive.it/it/privati/ego
[19] www.smart.com/it/it/index/app-center/smart-eq-control-app.html

- Pre-climatization;
- Notification in case of problems with tire pressure occur;
- Statistics on the expense of the car.

**BMW iPerformance**[20] is the BMW app dedicated to provide to users several functions like the possibility of opening and closing the car remotely, as well as the possibility of checking the car autonomy and the charging state. The app also offers a system of navigation that shows the available charging stations on the user's route. Moreover, the BMW iPerformance offers the possibility of pre-charging and pre-conditioning in order to have the car always warmed and fully charged. Another interesting characteristic is that BMW iPerformance offers the analysis of the efficiency in order to help the drivers obtain maximum autonomy. Another interesting work in progress feature is the planning of the trips considering public transportation, checking the timetable, the parking available and the possible combined option.

**MyFord mobile**[21] is the app created for several Focus Electric vehicles. This app was developed to help consumers easily manage vehicles functions. This app allows controlling the beginning and the end of charging, it shows the vehicle information (odometer, battery system health, tire pressure, etc.). Furthermore, the MyFord Mobile allows localising the vehicle and its closer available charging station. Finally, through the app the user will be able to pre-heat his/her car at a certain time, combining the nice sensation of a warm car, with the best conditions for a battery.

**Plug surfing**[22] is a system that offers several features for electric vehicle owners, such as the possibility to connect the drivers with all the charging stations in Europe through two ways:

- The charging key: holding the charging key to a charging station, the charging will start immediately;
- The Plugsurfing app that allows the user to find the closest charging point and to discover if it is available or occupied.

**Charge Point**[23] is a society that created several tools that can manage the EL-Vs charging. The application allows to find available stations, start charging using the phone, receive a notification when the car has finished charging or when a station opens, check station pictures in order to easily find the place (the pictures, together with other information about the stations are uploaded from the users), share real driver tips and schedule charging, set reminders and track the usage.

**Open Charge Map**[24] is a non-commercial non-profit service, supported by a community composed of interested parties around the world and developed by volunteers. The aim of this service is to develop and provide an open database of charging equipment. The benefit of Open Charge Map is that allows to share information openly, information coming from many locations and directly provided from the users.

[20] www.bmw.it/it/topics/mondo-bmw/bmw-connecteddrive/iPerformance.html
[21] play.google.com/store/apps/details?id=com.ford.mfm&hl=it
[22] www.plugsurfing.com/en/
[23] www.chargepoint.com/it-it/
[24] openchargemap.org/site

**e-go:** The service E-Go[25] proposed from the ALD automotive is based on the idea that renting a car (and in particular an electric car) is the best solution both for private and for business. For this reason, they created E-go, the 100% electric solution that allows drivers to move freely around the city (no LTZ for ELV-s, no restriction of any kind), spending less money and taking care of the environment. In order to offer more electromobility services, they developed also an application: My E-go that allows the users to better control their car, through several features as:

- Enel X Box Station: this feature allows the user to rent not just a car but also a box station to put in the users' private parking;
- A map with all the charging station available;
- Updated research and materials to always inform about the newest technology and the new ways of charge.

The overview of existing services presented in this chapter shows that there is already a large variety of services for electromobility and electric vehicles. Similarly to infomobility services for public transport, the main concern is the lack of integration of existing information. There are, for instance, several applications that provide information on the location of the charge point and other characteristics related to the type of plugs available or the real time availability. In this context, there is a high potential for NeMo to offer interesting conditions for the service providers to be part of the NeMo Hyper-Network and offer their services though the NeMo marketplace.

An overview of the presented existing services in relation to NeMo services is shown in Table 2.

| NeMo services | Existing services |
|---|---|
| NeMo Load forecasting due to EV charging | myEV electric vehicle logger & app<br>Fleetcarma |
| NeMo Thermal Battery Pre-conditioning | Tesla new mobile app update with battery pre-conditioning feature<br>BMW i3<br>Ford Global Technologies |
| NeMo Battery capacity calculation | Texas Instruments Battery Management System<br>Renesas Li-ion battery management solutions<br>Lithiumate Pro BMW<br>Electrip |
| EV Range Anxiety | Has To Be emobility products, PlugShare, Charge Hub calculator, NextCharge, MOVING, CHARGEBUMP, Charge Map, EVWAY (Route220), e-Go ADL automotive, Enel X Box Station, EQ Control, BMW iPerformance, MyFord mobile, Plug surfing, Charge Point, Open Charge Map |

Table 2: Overview of existing e-mobility services similar to NeMo services

[25] mobilitysolutions.aldautomotive.it/it/privati/ego

# 4. Definition of quality standards for NeMo services

Based on the analysis of the existing literature as previously presented, the present Chapter 4 provides a qualitative description of each NeMo service with the final objective of identifying the characteristics of data and services that are able to ensure an acceptable level of service quality. For this purpose, a template was filled in for each service. This includes the description of the type of service provided, the indicators of service quality, the indication of the failure outcome reported in the D1.1 Consolidated Version of the Use Cases and Actors' Requirements, the indicators of service quality and the type of input or output data, including the reference to D9.2 Quality Management Plan.

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | This field contains a brief description of the NeMo service |
| **Quality of service indicators** | This field presents the list of indicators to assess the quality of the service. The identification of the quality indicators is based on the literature review analysis. |
| **Measure of failure scenario** | This field is based on the failure outcome reported for each Use Case in D1.1. |
| **Minimum service quality level requirements / indicators** | This field reports for each of the identified quality measure, the minimum quality level to be fulfilled |
| **Data in** | This field describes the type of data that are deployed to provide the service |
| **Quality of Data in indicators** | This field describes the indicators for data quality based on the previous analysis of literature |
| **Data out** | This field describes the information provided to the final user of the NeMo service |
| **Quality of Data out indicators** | This field describes the indicators to assess the quality of information provided to the end user of the NeMo service |
| **Type of data collected from D9.2** | This field reports the data listed in D9.2 that are relevant to this service |

Table 3: Template for the description of data and service quality

## 4.1 Services provided by MOSAIC – BARCELONA Horizontal services

The horizontal services tested in Barcelona's test site are the Service brokerage, CPO monitoring and profiling, EV driver monitor and profiler and the Application service finder. These services are developed on top of the Nemo Hyper-Network in order to provide functionalities in the whole environment.

The use cases involved in the horizontal services tested in Barcelona are: UC03, UC49, UC50, UC51, UC54.

### 4.1.1 Service Brokerage

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The brokerage service aims to assist end users choosing the right charging station in terms of availability and price, while avoiding system congestion due to high demand of specific resources |
| **Quality of service indicators** | Accuracy, Consistency |
| **Measure of failure scenario** | Percentage of service requests that have been rejected (no response) |
| **Minimum service quality level requirements / indicators** | Response time of the service: 10 seconds<br>Correctness of the CP status and availability: 80%<br>Percentage of useful suggestions for the EV driver<br>Balance of the system load |
| **Data in** | Availability forecast of CP<br>Recurrent Places of an EV driver |
| **Quality of Data in indicators** | Accuracy of availability forecast of CP<br>Accuracy of the recurrent places<br>Accuracy of the recurrent schedules<br>Accuracy of recurrent trips |
| **Data out** | Suggestion of CPs to be used |
| **Quality of Data out indicators** | Accuracy of the suggestions |
| **Type of data collected from D9.2** | History of users' traveling, charging and destinations |

Table 4: Service Brokerage

### 4.1.2 EV driver monitor and profiling

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The EV driver monitor and profiling collects data from EV drivers in order to profile them in terms of charging and driving preferences. |
| **Quality of service indicators** | Accuracy, Completeness |
| **Measure of failure scenario** | Monitor deactivated, the activity is not monitored<br>Actor profile modifications not saved properly |

| | |
|---|---|
| | The user profile does not open or does not show any information |
| **Minimum service quality level requirements / indicators** | Response time of the service: 5 seconds<br>Accuracy of the geolocation: 15 m<br>Percentage of user coordinates tracked >70% |
| **Data in** | Track<br>EV driver preferences and contact information |
| **Quality of Data in indicators** | Percentage of user mobility path tracked<br>Percentage of user information |
| **Data out** | EV driver profiler |
| **Quality of Data out indicators** | Accuracy of the charging preferences<br>Accuracy of the recurrent places<br>Accuracy of the recurrent schedules<br>Accuracy of recurrent trips |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour User's trip plans |

Table 5: EV monitoring and profiling

Virtual sensors are software sensors that provide indirect measurements of abstract conditions by providing heterogeneous data. They can do computation and data elaborations. They are used in different fields to estimate the value of certain parameters or derive new information on the collected data. In the electromobility field, they can provide innovative and value-added services. They can support the management of the battery health, help the driver to select the charge point according to their preferences and also charging status. They are also supporting the trip itinerary planning by providing additional information.

### 4.1.3 CPO monitoring and profiling

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The CPO monitoring and profiling generates a profile for each CPO to predict the status of each charging point at any time between the current moment and the following week |
| **Quality of service indicators** | Accuracy |
| **Measure of failure scenario** | Monitor deactivated, the activity is not monitored<br>Actor profile modifications not saved properly<br>The CPO profile does not open or does not show any information |

| | |
|---|---|
| **Minimum service quality level requirements / indicators** | Response time of the service: 3 seconds<br>Accuracy of the predicted CP availability:<br>- next hour – Average accuracy > 90%<br>- next week – Average accuracy > 80% |
| **Data in** | CPO data |
| **Quality of Data in indicators** | Accuracy of CP information |
| **Data out** | CPO profile<br>Availability forecast of charging points |
| **Quality of Data out indicators** | Accuracy of the predicted availability of the CP |
| **Type of data collected from D9.2** | Charge infrastructure location and status |

Table 6: CPO Monitoring and Profiling

### 4.1.4 Application level finder

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The application level finder searches for services that can provide information regarding charging points / charging stations, extracts relevant information and provides it in a unified way |
| **Quality of service indicators** | Accuracy, Timeliness |
| **Measure of failure scenario** | The service doesn't return a successful response |
| **Minimum service quality level requirements / indicators** | Response time of the service: 5 seconds |
| **Data in** | Bounding Box<br>Charging connectors<br>Charging solutions<br>Payment methods |
| **Quality of Data in indicators** | Accuracy of CP information (charging connectors, charging mode, payment methods,…) |
| **Data out** | Charging pools |
| **Quality of Data out indicators** | Accuracy of CP information (charging connectors, charging mode, payment methods, CP location,…) |
| **Type of data collected from D9.2** | Charging infrastructure |

Table 7: Application Level Finder

## 4.2 Virtual Sensors provided as NeMo services

In NeMo Hyper-Network, the virtual sensors (VS) are seen as any other service hosted in a NeMo node. In the first phase, the VS gather data from the data provider, and internal algorithms compute and update indirect measurements. The output is given back to the users of the service, who are asking for this service.

### 4.2.1 Personal Mobility Probability

ICOOR (POLIBA) provides the Personal Mobility Probability VS that elaborates data about the historic trip of the given driver, and returns a set of possible routes for the next day, each one composed by the interpolation of several Point of Interests (POIs) described by the several parameters such as geocoordinates, arrival/departure timestamps, arrival/departure battery SoC.

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The service uses statistical algorithms to elaborate past trips history data of a given driver and derive the driver most probable routes during the next calendar day. Each route is a spatial-temporal path composed by the interpolation of Point of Interests (POI) such as the Start point (SP), the Charge/other intermediate stops and the End point (EP) of the trip. |
| **Quality of service indicators** | Average response time<br>Availability rate<br>Average user feedback: vote on the service usefulness from 1 (min) to 5 (max). |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 3000ms<br>Min availability rate: 98%<br>Average user feedback: 3 |
| **Data in** | Driver past trips data with POIs coordinates and EV backend data [velocity (km/h), distances (km), SoC, EV on/off status]. |
| **Quality of Data in indicators** | Accuracy of EV backend data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | Most probable route POIs. Each POI described by: POI latitude, POI longitude, arrival timestamp, departure timestamp, arrival SoC, departure SoC. |
| **Quality of Data out indicators** | Average response time<br>Availability rate<br>Average user feedback: vote on the service usefulness from 1 (min) to 5 (max). |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour |

| | |
|---|---|
| | History of users' traveling, charging and destinations<br>User's trip plans<br>Test users' background data (for example gender, age, preferences) |

Table 8: Personal Mobility Probability

### 4.2.2 Charge Price Prediction

The second VS by ICOOR (POLIBA) is the Charge Price Prediction that provides information on the Charge stations in the area of the driver, during the trip, and predicts the charge costs of the CPs that are selected by the driver. Thus, the VS provides CP static and dynamic information, including the stats of the CP and the cost for the driver.

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The service provides information about charge stations related to a specific time horizon (e.g. next 24 hours) in the area of interest (along the trip) of a given driver. Moreover, it predicts charge session cost for the given driver selecting specific charge point in the output list. |
| **Quality of service indicators** | Average response time<br>Availability rate<br>Average user feedback: vote on the service usefulness from 1 (min) to 5 (max). |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 3000ms<br>Min availability rate: 98%<br>Average user feedback: 3 |
| **Data in** | Charge point (CP) data: location; power; connection type; tariff; charge detail record; payment method; booking/reservation info.<br>EV/driver data: Personal/Vehicle mobility need, EV position; residual charge; desired charge. |
| **Quality of Data in indicators** | Accuracy of CPO and EV backend data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | For each charge point provides: location; power; connection type; tariff; payment method; booking/reservation possibility; predicted charge cost. |
| **Quality of Data out indicators** | Average response time<br>Availability rate |

| | |
|---|---|
| | Average user feedback: vote on the service usefulness from 1 (min) to 5 (max). |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour<br>History of users' traveling, charging and destinations |

Table 9: Charge Price Prediction

The following six VS by FKA are all battery-related services. The provide information or control requests affecting the battery or the battery management system.

### 4.2.3 AdaptiveSoCEstimation

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Aims to optimize the charging strategy such, that an optimum on vehicle usability and intended vehicle use is achieved. This mainly focuses to avoid charging the EV battery to 100% SoC always, which enhances the short-term usability but reduces battery lifetime. The service respects mobility demands and driver as well as vehicle characteristics to estimate the required mission SoC. The service only provides a proposal to the battery. The EV and the BMS can agree or overwrite the proposal. |
| **Quality of service indicators** | Average response time<br>Availability rate<br>Accuracy percentage of service output: correctness of output regarding vehicle usability<br>Accuracy percentage of service output: correctness of output regarding SoC overestimate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 10000ms (long term planning data)<br>Min availability rate: 98%<br>Accuracy percentage of service output: delta SoC underestimate<br>Accuracy percentage of service output: delta SoC overestimate |
| **Data in** | EV/driver data: Personal/Vehicle mobility need<br>EV/driver data: mobility behaviour<br>Environmental (optional, complementary): weather conditions |
| **Quality of Data in indicators** | Accuracy of EV backend data<br>Responsiveness of the Reporting Server (request time) |

| | |
|---|---|
| | Availability of the Reporting Server (percentage of online time) |
| **Data out** | Charging proposal with SoC over time characteristic |
| **Quality of Data out indicators** | Accuracy percentage of forecasted SoC information reflected to real world SoC demand. |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour<br>History of users' traveling, charging and destinations<br>Battery and vehicle information |

Table 10: Adaptive SoC Estimation

### 4.2.4 ImprovedSoCEstimation

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Estimate the status of charge (SoC) of an EV battery based upon the charging data as well as the mobility data of the vehicle. The system aims to respect the SoH of the battery. The service offers predictive SoC calculation thereby, useful for charging strategy alignment. |
| **Quality of service indicators** | Average response time<br>Availability rate<br>Accuracy percentage of service output: correctness of SoC estimate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 10000ms (long term planning data)<br>Min availability rate: 98%<br>Accuracy percentage of service output: delta SoC misalignment |
| **Data in** | EV: vehicle status, especially SoC and SoH of battery<br>EV/driver data: Personal/Vehicle mobility need<br>EV/driver data: mobility behaviour<br>Environmental (optional, complementary): weather and/or traffic conditions |
| **Quality of Data in indicators** | Accuracy of EV backend data<br>Accuracy and availability of environmental data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | SoC estimation (time and location based) |
| **Quality of Data out indicators** | Accuracy percentage of forecasted SoC information reflected to real world SoC. |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour<br>History of users' traveling, charging and destinations |

| | |
|---|---|
| | Battery and vehicle information |

Table 11: Improved SoC Estimation

### 4.2.5 BatteryCapacityCalculation

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Based upon the charging data and the mobility data correlated to the SoC, the state of health (SoH) of the battery of each EV will be estimated. The SoH is an expression of the maximum energy charging capacity of a battery and the accessible charging capacity. Knowing the behaviour of the SoH shall support services optimizing the battery lifetime and enhance as well the usability estimation of the vehicle. |
| **Quality of service indicators** | Average response time<br>Availability rate<br>Accuracy percentage of service output: correctness of SoC estimate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 60000ms (long term planning data)<br>Min availability rate: 98%<br>Accuracy percentage of service output: delta SoH misalignment by balancing of charged energy in comparison to SoC behaviour |
| **Data in** | EV: charging data<br>Environmental (optional, complementary): Weather conditions |
| **Quality of Data in indicators** | Accuracy of charging backend data (completeness)<br>Accuracy and availability of environmental data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | SoH estimation |
| **Quality of Data out indicators** | Accuracy percentage of forecasted SoH information reflected to real world SoH derived from charging data and reported SoC. |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour<br>History of users' traveling, charging and destinations<br>Battery and vehicle information |

Table 12: Battery Capacity Calculation

### 4.2.6 RemoteBMSParametrization

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The service collects field information about specific battery types from various EVs and their operation. By analysis of the collected data, the service proposes an optimized and individualized parameterization for the BMS, mainly the parameters of the utilized battery models, but furthermore the charging of balancing strategy, if these can be influenced. The parameter update then will be performed via NeMo. |
| **Quality of service indicators** | Average response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 60000ms (long term planning data)<br>Min availability rate: 98% |
| **Data in** | EV: charging data<br>EV: SoH and SoC behaviour<br>Driver/EV: personal/vehicle mobility demands<br>Environmental (optional, complementary): weather conditions |
| **Quality of Data in indicators** | Accuracy of charging backend data (completeness)<br>Accuracy and availability of environmental data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | BMS operation parameters; depending on battery type and accessible parameterization interface |
| **Quality of Data out indicators** | Completeness of BMS parameters<br>Statistical certainty indicator per parameter (deviation estimation) |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour<br>History of users' traveling, charging and destinations<br>Battery and vehicle information |

Table 13: Remote BMS Parametrisation

### 4.2.7 BatteryLoadManagement

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Service manages optimization of load and operation conditions of the battery of an EV. Mainly it aims to avoid the battery staying at 100% SoC for longer time by programming an optimized charging profile or by applying optimized discharging strategies during operation by load prioritisation. |
| **Quality of service indicators** | Average response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 5000ms<br>Min availability rate: 98% |
| **Data in** | EV: charging data<br>EV: discharging/load data<br>EV: SoH and SoC behaviour<br>Driver/EV: personal/vehicle mobility demands<br>Environmental (optional, complementary): Weather conditions |
| **Quality of Data in indicators** | Accuracy of charging backend data (completeness)<br>Accuracy and availability of environmental data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | vehicle charge proposal<br>electric load proposal, e.g., prioritisation of electric power demands |
| **Quality of Data out indicators** | - none - |
| **Type of data collected from D9.2** | Data relevant to actors' monitoring and profiling<br>Data relevant to vehicles' location and charging behaviour<br>History of users' traveling, charging and destinations<br>Battery and vehicle information |

Table 14: Battery Load Management

### 4.2.8 ThermalBatteryPreconditioning

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Based upon estimated drive-off times, the service improves battery performance by initiating a thermal preconditioning of the vehicle before drive-off. The service continuously is able to adapt the thermal precondition planning by observing changes on the allocated mobility needs as well as by respecting driver preferences and environmental conditions (optional). |
| **Quality of service indicators** | Average response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max response time: 5000ms<br>Min availability rate: 98% |
| **Data in** | Driver/EV: personal/vehicle mobility demands<br>Environmental (optional, complementary): Weather conditions |
| **Quality of Data in indicators** | Accuracy of mobility backend data (completeness, correctness)<br>Accuracy and availability of environmental data<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | Preconditioning proposal |
| **Quality of Data out indicators** | Accuracy percentage of required in-time forecasted precondition request and non-in-time preconditioning requests (multidimensional as “to early”, “to late”, “non needed”) |
| **Type of data collected from D9.2** | Data relevant to actors’ monitoring and profiling<br>Data relevant to vehicles’ location and charging behaviour<br>History of users’ traveling, charging and destinations<br>Battery and vehicle information |

Table 15: Thermal Battery Preconditioning

## 4.3 Grid services: MicroGridInfo by Verbund

Access to EV data will be done via a Gireve service that can access EVs through an interface provided by Vehicle Manufacturer(s). Smatrics is already publishing charge point data in near-real-time and they are available through another Gireve NeMo service, which has access to CP data from the Hubject platform based on the NeMo Inter-roaming protocol.

**MicroGridInfo**

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The service MicroGridInfo provides near-real-time information about high-power charging stations that will be commissioned within the SYNERG-E project (Verbund and Smatrics are partners in this project) in Austria and Germany.<br>Metrics include static (“hard”) power/energy limits of the local micro-grid as well as dynamic limits depending on the current mode of operation of the micro-grid management system as well as the state of the local battery (state of health, status of charge, …).<br>Furthermore, a CPO can demand a forecast (estimate) of the cost of a given future (near-time) load profile at the charge point. |
| **Quality of service indicators** | • Availability<br>Percentage of service online time<br>• Accuracy of price predictions:<br>relative deviation of predicted CPO-side costs versus actual costs on end-of-month bill.<br>• Scope of service:<br>Number of Charge Points covered by the service |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Currently, the service is only used as a demo (Smatrics will not actually use price forecasts for dynamic B2C pricing, etc.), therefore requirements are currently low. Once the service is actually used, required quality levels must rise.<br>• Required availability:<br>Currently: > 0<br>When actually used: > 98%<br>• Required accuracy of price predictions:<br>Currently: Deviation < 1000%<br>When actually used: Deviation < 20%<br>• Required scope of service:<br>Currently: At least 1 charge point (for demo)<br>When actually used: At least ~3 charge points |
| **Data in** | The service MicroGridInfo retrieves all required data from the “SYNERG-E Reporting Server” and or the “SYNERG-E Time-Series Database”, in particular:<br>• Via the Reporting Server:<br>○ Meta-data about locations<br>○ Applicable energy/grid tariffs per location<br>○ Price calculations for the current billing period |

| | |
|---|---|
| | • Via the Time-Series Database:<br>  ○ Current metrics of the local battery<br>  ○ Current mode, settings, and metrics of the micro-grid management system |
| **Quality of Data in indicators** | Time resolution of Time-Series data (for static and dynamic metrics about the battery and the micro-grid management system)<br>Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | Output metrics of the service are:<br>maximum available power from the grid<br>maximum available additional power from the local battery<br>capacity (maximum energy content) of the local battery<br>current charge of the local battery<br>current settings of the SYNERG-E micro-grid management system (peak shaving, grid services)<br>“grid forecast” for a given future (near-time) load profile:<br>relative cost of the given load profile (estimate)<br>expected status of charge of the local battery at the end of the charge of process |
| **Quality of Data out indicators** | Accuracy of price prediction (see above) |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services<br>Charging infrastructure |

Table 16: MicroGridInfo

## 4.4 Electromobility Actors’ Services

### 4.4.1 Trustee_getData service

The Trustee service proxy will be used in the Italian test site and it can be considered as a sort of external monitoring service and a proof of concept of a B2C model implemented in NeMo.

For what concerns the official use cases related to the BS2 that will be demonstrated in the Italian test site, all of them (UC04, UC11, UC12 and UC13) are associated to services that will be invoked by TomTom in order to provide to the driver a better (NeMo improved) navigation and booking system.

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The service Trustee_getData provides near-real-time telematic data for a specific clearance, namely the id of the contract double signed by the user for the provisioning of vehicle data to an external provider. |
| **Quality of service indicators** | Availability<br>Percentage of service online time<br>Responsiveness: |

| | |
|---|---|
| | measures the speed of the requests completion provided that the service has been invoked after the minimum time from the previous invocation as stated in the SLA. |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | The service will be used during the Italian test site with a fix SLA of a maximum invocation frequency of 0.1 Hz.<br>Required availability:<br>> 98%<br>Required accuracy of price predictions:<br>< 1 sec (calculated from the actual invocation of the atomic service and the delivery time of the result) |
| **Data in** | The service requires as REST parameter the clearance Id that is mapped by the OEM backend to a valid vehicle data. |
| **Quality of Data in indicators** | Valid/Invalid clearance Id |
| **Data out** | The service returns the last vehicle data available on the OEM backend as a json structure containing for each signal id:<br>• Name<br>• Value<br>• Unit<br>• Gps coordinates<br>• Timestamp |
| **Quality of Data out indicators** | Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Type of data collected from D9.2** | EV data |

Table 17: Trustee getData

### 4.4.2 GetEVSEData

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by a NeMo partner to get all EVSE data included in the scope contractualised with GIREVE platform. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response time : 3000ms<br>Min availability rate: 98% |

| | |
|---|---|
| **Data in** | Pagination information |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats) |
| **Data out** | EVSE information |
| **Quality of Data out indicators** | Completion rate for each EVSE |
| **Type of data collected from D9.2** | Charge infrastructure location and status |

Table 18: GetEVSE data

### 4.4.3 GetEVSEDynamicDataChanges

| **Service characteristics / level and data quality attributes** | **Description** |
|---|---|
| **High level description of the NeMo service** | Web service used by a NeMo partner to get new dynamic status of EVSEs that have changed since a given date.<br>These EVSEs are covered by the scope contractualised between the NeMo partner and GIREVE. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response time: 3000ms<br>Min availability rate: 98% |
| **Data in** | « dateFrom » and « dateTo » defining the requested period |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats |
| **Data out** | Dynamic status of EVSEs |
| **Quality of Data out indicators** | - |
| **Type of data collected from D9.2** | Charge infrastructure location and status |

Table 19: Get EVSE Dynamic Data Changes

### 4.4.4 GetEVSEStaticDataChanges

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by a NeMo partner to get new definition of EVSEs that have changed since a given date.<br>These EVSEs are covered by the scope contractualised between the NeMo partner and GIREVE. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response time: 3000ms<br>Min availability rate: 98% |
| **Data in** | « dateFrom » and « dateTo » defining the requested period |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats |
| **Data out** | EVSE information |
| **Quality of Data out indicators** | Completion rate for each EVSE |
| **Type of data collected from D9.2** | Charge infrastructure location and status |

Table 20: GetEVSE Data Changes

### 4.4.5 GetGeolocatedEVSEData

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by a NeMo partner to get all EVSE located in a given perimeter defined by geocoordinates and a round.<br>These EVSEs are covered by the scope contractualised between the NeMo partner and GIREVE. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |

| | |
|---|---|
| **Minimum service quality level requirements / indicators** | Max Response time: 3000ms<br>Min availability rate: 98% |
| **Data in** | Geocoordinates and filter information (connectorType, etc.) |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats) |
| **Data out** | EVSE information |
| **Quality of Data out indicators** | Data corresponding to the request (EVSEs in the requested area, compliant with filters) |
| **Type of data collected from D9.2** | Charge infrastructure location and status |

Table 21: GetGeolocatedEVSeData

### 4.4.6 RemoteStartChargeService characteristics / level and data quality attributes

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by an EMSP to start charge remotely on a charging point. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response time (in GIREVE system, exclude the time of the request to the CPO): 2000ms<br>Min availability rate: 98% |
| **Data in** | Driver authentication information. EVSE identifier |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats) |
| **Data out** | Authorisation result |
| **Quality of Data out indicators** | Rating of succeeded requests (Http return code 2xx) |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services |

Table 22: RemoteStartChargeService

### 4.4.7 RemoteActionRequest

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by an EMSP to send actions relative to a charging session. (Stop charge, …) |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response time (in GIREVE system, exclude the time of the request to the CPO): 2000ms<br>Min availability rate: 98% |
| **Data in** | Charge Session identifier, actionType (stop, pause, restart, etc.) |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats) |
| **Data out** | Action result |
| **Quality of Data out indicators** | Rating of succeeded requests (Http return code 2xx) |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services |

Table 23: Remote Action Request

### 4.4.8 UpdateWhiteList

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by an EMSP to update list of its RFID tokens accepted for a charge. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response: 2000ms<br>Min availability rate: 98% |

| | |
|---|---|
| **Data in** | Driver tokens information |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats) |
| **Data out** | Result of the insert |
| **Quality of Data out indicators** | Rating of succeeded requests (Http return code 2xx) |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services |

Table 24: UpdateWhiteList

### 4.4.9 GetChargeDetailRecords

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Web service used by an EMSP to get CDRs relative to charges of its customers on charging points of CPOs. |
| **Quality of service indicators** | Average Response time<br>Availability rate |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Max Response: 2000ms<br>Min availability rate: 98% |
| **Data in** | « dateFrom » and « dateTo » defining the requested period |
| **Quality of Data in indicators** | Compliance rate of requests (requests not rejected because of their formats) |
| **Data out** | Charge Detail Records |
| **Quality of Data out indicators** | Rating of succeeded requests (Http return code 2xx) |
| **Type of data collected from D9.2** | History of users' traveling, charging and destinations |

Table 25: GetChargeDetailRecords

### 4.4.10 Load Management

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | The load management service aims at performing smart charging of EV's according to grid and EV grid criteria |
| **Quality of service indicators** | Availability<br>Percentage of service online time<br>Accuracy of load management:<br>Ensuring that the overall capacity of a given charging site does not go over the actual availability of the site. Ensuring the EV energy requirements are met if there is enough charging energy at a given charging site |
| **Measure of failure scenario** | Percentage of service requests (calls) that cannot be fulfilled (return an error) |
| **Minimum service quality level requirements / indicators** | Availability: of service > 95%<br>Overall capacity of a charging station must not be lower than the overall amount of energy dispatched to EVS<br>EVs must be charged if there is remaining capacity in a given test site |
| **Data in** | CPO energy pricing<br>CPO locations<br>EV energy requirements<br>Via external data adaptors:<br>CPO energy pricing<br>CPO locations<br>EV energy requirements |
| **Quality of Data in indicators** | Responsiveness of the Reporting Server (request time)<br>Availability of the Reporting Server (percentage of online time) |
| **Data out** | Smart charging power schedule<br>Location of charging station where a given charge schedule could take place<br>Cost of charging operations at a given charging spot location and a given charging profile |
| **Quality of Data out indicators** | The charging profile must be such that energy requirements of the EV must fulfil charging energy requirements of an EV. |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services<br>Charging infrastructure |

Table 26: Load Management

### 4.4.11 Data translator configuration

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | This is a function offered by the Data Translator to the Registry, in practice through the NeMo Marketplace as part of the registration process of a new service. It allows the Service Provider to configure their local Translator instance so that said service can seamlessly interact according to the CIM.<br>During this procedure, the Data Translator is provided with the PSM-to-CIM mapping, which, expressed though the appropriate XML structure, reflects the exact relationships between the platform-specific data model based on which the service is implemented and the CIM, in both semantic and syntactic terms. Further, this mapping includes SP-defined policies for data handling, by allowing the SP to specify visibility levels for all data involved at a fine-grained level (expressed in CIM). Based on this information, the Data Translator internally generates the logic required to transform incoming requests and outgoing responses at invocation time (see following service below), expressed in the Transformation Workflow Language. |
| **Quality of service indicators** | Availability<br>Percentage of service online time<br>Error rate<br>Percentage of service requests (calls), i.e., configuration attempts, that cannot be fulfilled (return an error)<br>Delay<br>The elapsed time from when a configuration request<br>is first submitted to when the configuration result is produced.<br>Accuracy/correctness of translated messages:<br>Assessment of correctness and accuracy during message translation at runtime (see service described below), and by extension, availability of target service<br>Scope of service:<br>Percentage of offered services supported by the configuration functionality |
| **Measure of failure scenario** | Not applicable. |
| **Minimum service quality level requirements / indicators** | Intended values when actually used:<br>Required availability: > 99,999%<br>Acceptable error rate: 0%<br>Acceptable delay: < 50ms<br>Required scope of service: 100% of deployed services |
| **Data in** | The Data Translator configuration service requires:<br>the service identifier |

| | |
|---|---|
| | the URL of the appropriate mapping XML file, residing at the local Service Artefacts Server. |
| **Quality of Data in indicators** | Correctness (syntactic and semantic) of the mapping XML file<br>Responsiveness of the Service Artefacts Server (request time)<br>Availability of the Service Artefacts Server (percentage of online time) |
| **Data out** | The output of the service is the translation workflow, i.e., a description of well-specified steps that will guide the translation of CIM-based request messages to PSM requests and, conversely, PSM-based responses to CIM messages. This is kept internally and not provided to another user or component. |
| **Quality of Data out indicators** | Accuracy and correctness of translated messages at runtime and, by extension, successful target service delivery. |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services (mapping files coming from respective Service Providers) |

Table 27: Data Translator Configuration

## 4.5 Other functionalities evaluated for quality level

### 4.5.1 Data translation

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | This function is offered by the Data Translator to the Integration Bus of the same Node during the execution of any service offered by this Node through the Marketplace. It facilitates the orchestration of appropriate transformation functions on each service request/response at service invocation time, as specified during the configuration of the Service Provider's Data Translator (see above), so that available services can be offered to any consumer connected to the NeMo Hyper-Network on the basis of the CIM. |
| **Quality of service indicators** | Availability<br>Percentage of service online time<br>Error rate<br>Percentage of service requests (calls), i.e., message translation attempts, that cannot be fulfilled (return an error)<br>Delay<br>The elapsed time from when a translation request is first submitted to when the translated outcome is produced.<br>Accuracy/correctness of translated messages:<br>Assessment of correctness and accuracy of translated messages, and, by extension, availability of target service<br>Scope of service:<br>Percentage of offered services supported by the translation functionality |
| **Measure of failure scenario** | Not applicable. |

| | |
|---|---|
| **Minimum service quality level requirements / indicators** | Intended values when actually used:<br>Required availability: > 99,999%<br>Acceptable error rate: 0%<br>Acceptable delay: < 50ms<br>Required scope of service: 100% of deployed services |
| **Data in** | The Data Translator configuration service requires from the Integration Bus:<br>the service identifier<br>the message data to be translated<br>the desired output format (XML or JSON). |
| **Quality of Data in indicators** | This field describes the indicators for data quality based on the previous analysis of literature<br>Quality and correctness of request from Integration Bus |
| **Data out** | This field describes the information provided to the final user of the NeMo service<br>The output of the service is the translated message. More specifically, said function on the one hand receives CIM request message data and transforms them to PSM request message data, and, on the opposite direction, receives PSM response message data and outputs CIM response message data. |
| **Quality of Data out indicators** | This field describes the indicators to assess the quality of information provided to the end user of the NeMo service.<br>Accuracy and correctness of translated messages and, by extension, successful target service delivery. |
| **Type of data collected from D9.2** | This field reports the data listed in D9.2 that are relevant to this service<br>Data from connected electromobility actors and services (any data service requests and responses)<br>Potentially other more specific types of data from the D9.2 list that may fall under above category |

Table 28: Data Translator

### 4.5.2 Identity and access management services

| Service characteristics / level and data quality attributes | Description |
|---|---|
| **High level description of the NeMo service** | Component-to-Component Authorisation: This function enables each NeMo Node component to verify if another requesting component (represented by an access token) can perform an operation on a specific resource (i.e., on the requested component). It therefore performs access control at the API level among Node components. |
| **Quality of service indicators** | Availability<br>Percentage of service online time<br>Error rate |

| | |
|---|---|
| | Percentage of service requests (calls), i.e., authorisation attempts, that cannot be fulfilled (return an error)<br>Delay<br>The elapsed time from when an authorisation request is first submitted to when the authorisation decision is returned.<br>Correctness of authorisation decisions:<br>Assessment of its efficiency regarding component protection and access control |
| **Measure of failure scenario** | Not applicable. |
| **Minimum service quality level requirements / indicators** | Intended values when actually used:<br>Required availability: > 99,999%<br>Acceptable error rate: 0%<br>Acceptable delay: < 50ms |
| **Data in** | The Component-to-Component Authorisation receives from the serving component:<br>an authorisation request containing a client token (indicating who requests access)<br>the operation (indicating what kind of access is requested)<br>the requested resource (indicating the API to be accessed). |
| **Quality of Data in indicators** | Quality and correctness of authorisation request<br>Quality of access token retrieved by the authentication component (see below) |
| **Data out** | The output of the service is the authorisation evaluation result, indicating whether access is granted or not. |
| **Quality of Data out indicators** | Adequacy of component protection. |
| **Type of data collected from D9.2** | Data from connected electromobility actors and services (data concerning a component's resources to be protected)<br>User credentials from connected actors using the "Service Creation", "Service Execution", "Service Delivery" domains of the OPEN Cloud Marketplace |

Table 29: Identity and Access Management

# 5. Conformance processes for self-certification of data and services provision

The review on the methodologies and indicators to evaluate the quality of services from the previous sections has underlined the importance of having services for which the quality is continuously ensured. In this chapter, the self-certification methodology is presented as an alternative to enrol an external service certifier who takes care that the minimum level of data and service quality offered by the NeMo Hyper-Network and NeMo marketplace is met.

The methodology for self-certification is based on the identification of a common set of indicators based on the literature review analysis presented in Chapter 2, and on the description of service and data quality indicators for each NeMo service provided in Chapter 4. Since the NeMo services and those that will be created after the conclusion of the project have characteristics that can be evaluated only with specific indicators, the objective of this section is to select a list of high-level indicators that could be adapted and measured for each service in order to ensure quality.

Table 30 presents an overview of a selection of indicators for each NeMo service. The table shows that the same indicators are deployed to measure the quality of several services but with different threshold values.

| NeMo service | Response time | Availability (rate) | Accuracy |
|---|---|---|---|
| **Service Brokerage** | 10 seconds | 80% | x |
| **EV driver monitor and profiling** | 5 seconds | - | x |
| **CPO monitoring and profiling** | 3 seconds | - | x |
| **Application level finder** | - | - | x |
| **Personal Mobility Probability** | 3000ms | 98% | x |
| **Charge Price Prediction** | 3000ms | 98% | x |
| **AdaptiveSoCEstimation** | 10000ms | 98% | x |
| **ImprovedSoCEstimation** | 10000ms | 98% | x |
| **BatteryCapacityCalculation** | 60000ms | 98% | x |
| **RemoteBMSParametrization** | 60000ms | 98% | - |
| **BatteryLoadManagement** | 5000ms | 98% | - |
| **ThermalBatteryPreconditioning** | 5000ms | 98% | x |
| **MicroGridInfo** | - | 98% | x |

| | | | |
|---|---|---|---|
| **Trustee_getData service** | - | 98% | x |
| **GetEVSEData** | 3000ms | 98% | - |
| **GetEVSEDynamicDataChanges** | 3000ms | 98% | - |
| **GetEVSEStaticDataChanges** | 3000ms | 98% | - |
| **GetGeolocatedEVSEData** | 3000ms | 98% | - |
| **RemoteStartCharge** | 2000ms | 98% | Rating of succeeded requests (Http return code 2xx) |
| **RemoteActionRequest** | 2000ms | 98% | Rating of succeeded requests (Http return code 2xx) |
| **UpdateWhiteList** | 2000ms | 98% | Rating of succeeded requests |
| **GetChargeDetailRecords** | 2000ms | 98% | Rating of succeeded requests |
| **Load Management** | - | 95% | x |
| **Data translator configuration** | - | 99,999% | x |
| **Data translation** | - | 99,999% | x |

Table 30: Summary of the quality indicators of NeMo services

Based on the summary table, in order to measure the quality of all existing NeMo services, three indicators of service and data quality can be identified: **Response time**, **Accuracy** and **Availability**. These three indicators are consistent with the ones identified in the literature review analysis in which Timeliness, Accuracy and Accessibility are the categories of indicators mostly cited by the considered authors. With reference to the quality of service, a feedback provided by the user on the quality of the service offered could be asked to evaluate **efficiency**, **system availability**, **reliability** and **easiness to use**.

Self-certification of service and data quality could be the result of two parallel activities: a self and continuous evaluation of the service in terms of Response Time, Accuracy and Accessibility; an evaluation obtained by the user on the service efficiency, availability, reliability and easiness to use.

# 6. Business model for service certifier

As mentioned in the introduction NeMo has created a Hyper-Network, a technical environment which enables the electromobility stakeholders to interconnect among themselves to create added value services for the end customers, the EV drivers. In order for this Hyper-Network, under the future governance of the BAEM (Business Alliance for Electromobility), to be attractive, different tools for service providers and consumers need to be provided to assure a good quality of the services published.

Therefore, the role of the service certifier was introduced. This actor will have different roles:

- Assure, for the sake of service providers or directly for the BAEM, that the services published on the Hyper-Network are fully compliant with the level of quality we have defined in the previous sections.
- On a regular basis, be able to screen the concerned services to control if the quality of services is still coherent with the Hyper-Network requirements.

Within the process of service creation described below, the service certifier will have to verify the compliance with the quality requirements:




Then, when the service is published and running, the service certifier will have the possibility, if this is required by the service provider or the BAEM, to do a regular check of the quality.

In addition to verification of quality compliance, the service certifier can also have the role of mapping the services and support the service provider in publishing their services, as it is shown below:




The service certifier will be supported either by the service providers who are willing to have an external control of their service, or directly by the BAEM for quality screening of the different services published on the Hyper-Network:

- Service certifier under the BAEM: This can be implemented via a regular sub-contract by the managing organization of the NeMo Hyper-Network. The goal of such contract is for the BAEM to have a clear overview of the quality of services published

- Service certifier under the service provider(s): For service quality guarantee, the service provider will be able to conclude a contract with the service certifier. The financial terms, and the guarantee provided by the service certifier are to be set by the parties.

This approach will be optional for the service providers while the BAEM will devote all its efforts to facilitate the customer journey of service providers in the service creation process by providing clear documentation.

| Data Quality | Service Quality |
|---|---|
| efficiency | response time |
| system availability | availability |
| reliability | accuracy |
| easiness to use | - |

Table 31: Selected data and service quality indicators

Another business model for a service certifier can be based on a different labels system such as Trip Advisor. In this mode, the services are rated by service consumers and the best services are high-ranked. The service certifier would control and verify the veracity of the rating and could propose to the service provider an action plan to increase its service quality.

# 7. Methodology for global adoption of NeMo requirements and specifications

This chapter describes and defines the process to enrol new actors in NeMo. Indeed, the Hyper-Network will have a significant impact to ease the EV mass market adoption only if there is a significant share of EV ecosystem stakeholders connected to it.

The objective is to ease as much as possible the access of new actors in the NeMo Hyper-Network. This section defines how to attract new members, the marketing processes and what is the added value for members who are taking part in the BAEM.

From a marketing point view, before a business partner technically sets up a NeMo Node or gets hosted by a Node operator, the potential members will be able to visit a website in order to visualize the services available on the network before further engaging with the BAEM. The website will have the name of “Business Alliance for electro-mobility” and there the service providers will be able to publish, on a business level, the services they have developed and show them in a publicly accessible area of the website.

## 7.1 Landing page

The website could have the same structure as the ITS Observatory site (https://www.its-observatory.eu/) but the graphical chart of the actual NeMo website.




A first part will be the presentation of the BAEM. It should content:

- Description of the BAEM
- Explanation of challenges
- Explanation of the genesis of the network
- Description of the service providers / consumers attributes
- Description of the benefits to join the Hyper-Network
- Explanation of the enrolment process
- Description of the Terms & Condition for joining

## 7.2 Business Marketplace

The part of the Business Marketplace will allow service providers to list and describe the services they have developed (or are currently developing). This service publication will be free of charge but will require the creation of an account.

To publish its service, the provider will have to fill in a standard form after creating his/her account. In particular, he/she will have to choose if the services will be published only on the business marketplace or made available through the Hyper-Network. The services that the company will choose to expose will be visible only by service consumers belonging to the target group of the service provider.

In case the services are not published through the Hyper-Network, this will require no fee. Otherwise, a yearly fee, to be determined, will be charged for the technical usage of the Hyper-Network, for covering administrative cost and running cost. This fee will be in addition to the cost to implement a NeMo Node, or the cost to be hosted by a Node Operator (Affiliate partner business model depends on the business arrangement between respective business partners.).

Service consumers connecting to the BAEM website, will be able to visualize the services available on the marketplace and the ones accessible via the Hyper-Network. Without creating a BAEM account (connecting technically to the network), they will not be able to see the detailed description of the services, neither the business contact of the service provider.

Similarly to service providers, the service consumers will be able to choose between the business marketplace and the Hyper-Network access, having two levels of fees. It is important to note that if a service provider / consumer subscribes to the Hyper-Network, he will be able to view the services listed on the business marketplace and have access to those technically published to the Hyper-Network. To sum up, the table below describes the different rights among service consumers, providers and website visitors.

| | Service providers | | Service consumers | | ALL |
|---|---|---|---|---|---|
| | **Marketplace** | **Hyper-Network** | **Marketplace** | **Hyper-Network** | **BAEM Website** |
| **Account creation** | YES | YES | YES | YES | NO |
| **Access to the Hyper-Network** | NO | YES | NO | YES | NO |
| **Visualization of services available** | NO | YES | YES | YES | YES (but business contact & Detailed description hidden) |
| **Fee** | NO | LOW | LOW | MEDIUM | NO |
| **Support from BAEM** | NO | YES | NO | YES | NO |

Table 32: Usage rights per service consumer, provider and visitors

## 7.3 Hyper-Network customer journey

For this part, the service provider and service consumers are distinguished, even if the process is quite the same. In general, the only access point to further engage with the Hyper-Network is through the BAEM website.

### 7.3.1 Service providers

When the service provider would like to access to the Hyper-Network and publish a service, he will have to:

- Create a BAEM account (not mandatory if he already publishes a service through the market-place)
- Set a NeMo Node or be hosted by a Node Operator. All necessary documentation will be at his disposal
- Download the Hyper-Network components for service providers (service creation tool, service registry, translator, service delivery) as described in the WP4
- Develop his service and map it with the CIM (at this point he will be able to contact the BAEM support by email)
- Provide the service offering for service consumers
- Publish his service

Here below some screenshots of the already existing access page to the Hyper-Network.

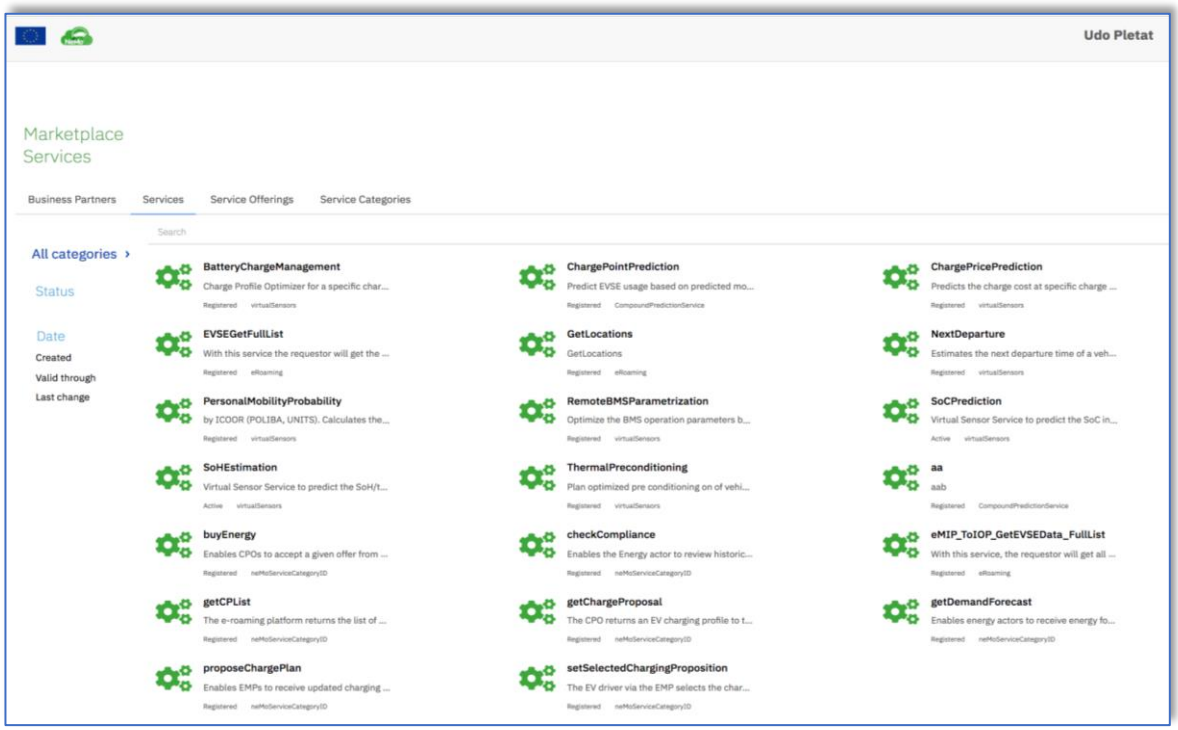



Figure 6: Access page to the NeMo Hyper-Network - Marketplace

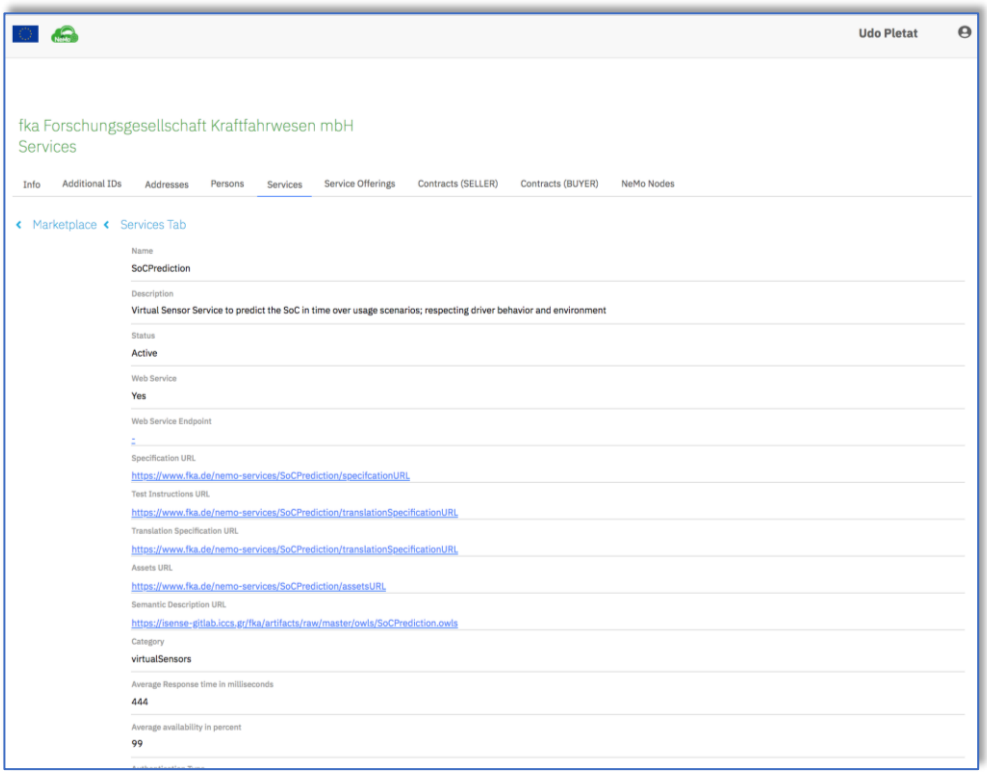



Figure 7: Access page to the NeMo Hyper-Network - Marketplace services

### 7.3.2 Service consumers

When the service consumer would like to access to the Hyper-Network and publish a service, he will have to:

- Create a BAEM account (not mandatory if he has already subscribed to the Marketplace access);
- Set a NeMo Node or be hosted by a Node Operator. All necessary documentation will be at his disposal;
- Download the Hyper-Network components for service consumers (service registry, translator, service delivery) as described in the WP4;
- Contract the service of interest with the concerned provider;
- Consume the contracted service according to the terms set by the provider.

To conclude on the methodology, the easiest customer journey for service providers and consumers needs to be as attractive as possible. Another very important aspect is, of course, the BAEM fee to be charged to service providers and consumers. No or low fixed fees, at least at the beginning of the life of the Hyper-Network, are recommended to attract the maximum number of users.

# 8. Conclusion

This Deliverable presents a methodology to ensure the quality of electromobility services offered by NeMo. Based on the analysis of existing studies and their comparison, that data quality should be measured based on the following criteria: Timeliness, Accuracy and Accessibility. The identification of the data quality indicators for each service finally confirmed that **Response Time, Availability** and **Accuracy** are the most appropriate values to be measured for assessing NeMo services data quality level. The brief analysis on existing methodologies to measure service quality instead shows that **efficiency**, **system availability**, **reliability** and **easiness to use** could be considered as the most appropriate criteria for the evaluation of the service quality by end users.

The first part of the deliverable presents existing electromobility services. It is clear that several functionalities overlap so that, for instance, different services offer the same information (charge point location, route planning for an EV), which is though fragmented per region or provider. In this context, NeMo represents a huge opportunity for connecting and integrating the services in one single environment. The involvement of existing electromobility service providers into the Business Alliance for Electromobility (BAEM) should be pursued in the future.

In the last part of the deliverable, the self-evaluation methodology is presented as an approach that consists of two parallel phases: a self-evaluation of the service provider based on the data quality indicators (Response Time, Availability and Accuracy) and an end-user's evaluation in the form of a feedback provided in terms of service quality indicators (efficiency, system availability, reliability and easiness to use). The business model for an external certifier is also presented. This Actor will have the task to check and ensure service quality and will be contracted either by the service provider or by the BAEM.

Finally, the last chapter presents a strategy for the adoption of NeMo. It uses a website, where the BAEM is presented and accessed, and a Business Marketplace where the service providers list their products.

Overall, the more actors connected to the Hyper-Network, the more business opportunities will be created. The NeMo Hyper-Network will allow the connection among actors located in different European countries while thanks to the Common Information Models, standardized data provision will be feasible. For the EV data, for instance, it will be easier to develop more sophisticated services. Added value services, such as the itinerary planning, which are in need of a lot and various data sources (information on topology, other EV users' charging, weather, etc.), will greatly benefit from the availability of all this data and services in the Hyper-Network.

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**For more information:**

NeMo Project Coordinator:
**Dr. Angelos Amditis**,
Research Director
Institute of Communication & Computer Systems (ICCS)
*9 Iroon Politechniou Str.*

*GR-157 73 Zografou, Greece*

a.amditis@iccs.gr
http://nemo-emobility.eu

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